Nasir Yasin

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Department of Mathematics and Statistics, Old Dominion University, Norfolk, VA 23529, USA

LinkedIn | Google Scholar | ORCID | GitHub

Research Interests

Machine Learning; Nonparametric and Total Variation Regression; Numerical Analysis; Scientific Computing; Computational Fluid Dynamics; Data-Driven Modeling.

Education

Ph.D. in Computational and Applied Mathematics 2023 – Present
Concentration: Mathematics of Data Science
Old Dominion University Norfolk, VA, United States

- **Proposed Dissertation:** *Advancing Robust and Interpretable Nonparametric Total Variation Regression for Complex Multivariate Process Data*

Master of Science (M.S.) in Computational and Applied Mathematics 2023 – 2025
Concentration: Applied Mathematics
Old Dominion University Norfolk, VA, United States

- **Thesis:** *Entropy and Gradient Stable Scheme for Hyperbolic Burgers Equation*

Master of Science (M.S.) in Mathematics 2019 – 2021
COMSATS University Islamabad, Islamabad Campus Islamabad, Pakistan

- **Thesis:** *Flow Features and Force Statistics Analysis of Flow Past Two Offset Cylinders*

Bachelor of Science (B.S.) in Mathematics 2014 – 2018
International Islamic University Islamabad Islamabad, Pakistan

- **Academic Focus:** Foundations of Pure, Applied, and Computational Mathematics

Experience

Graduate Assistant Aug 2023 – Present
Department of Mathematics and Statistics, Old Dominion University Norfolk, VA, United States

- Conduct research on robust and interpretable nonparametric total variation regression methods for complex multivariate process data through mathematical analysis, algorithm development, numerical experimentation, and predictive modeling.
- Support undergraduate mathematics instruction through classroom assistance, tutoring, supplemental instruction, and student mentoring, including the SMART Room, Applied Calculus Lab, and online tutoring.

Research Assistant Apr 2022 – Aug 2023
Applied Fluid Laboratory, Kyungpook National University Daegu, South Korea

- Contributed to the National Research Foundation of Korea (NRF)-funded project titled *Improved Energy Recovery Performance of Oscillated Wings Using Articulated and Flexible Bodies*, focusing on flow dynamics and energy-harvesting performance.
- Investigated unsteady flow around stationary, oscillating, articulated, and flexible rotating bodies across different Reynolds numbers and geometric configurations, analyzing vortex shedding, wake interactions, hydrodynamic forces, and energy-conversion efficiency.
- Implemented single- and multiple-relaxation-time lattice Boltzmann methods (SRT-LBM and MRT-LBM) coupled with large-eddy simulation to model incompressible flows and high-Reynolds-number wake behavior.
- Developed and optimized GPU-accelerated parallel simulation codes in C++ using CUDA to improve computational efficiency in large-scale numerical simulations.
- Post-processed numerical results using Tecplot and MATLAB to generate vorticity contours, streamline plots, force histories, frequency spectra, and publication-quality figures for technical reports and research manuscripts.

Research Assistant and Graduate Researcher Jan 2020 – Apr 2022
COMSATS University Islamabad Islamabad, Pakistan

- Served as a Research Assistant on a project funded by the Higher Education Commission (HEC) of Pakistan under the National Research Program for Universities (NRPU), titled *Numerical Study of Wake and Thrust Around Multiple Cylinders at Moderate Reynolds Numbers in Various Configurations*.
- Conducted graduate research from Feb 2021 to Mar 2022 in computational fluid dynamics, numerical modeling, and the evaluation of numerical methods and research gaps.
- Investigated flow behavior around multiple bluff bodies at moderate Reynolds numbers under different geometric configurations and spacing conditions.
- Analyzed flow separation, steady-to-unsteady transitions, wake structures, vortex shedding, and drag and lift force statistics.
- Implemented the lattice Boltzmann method and developed Fortran programs for numerical simulation and data analysis.
- Post-processed computational results and prepared plots, technical figures, and comparative analyses of drag, lift, and wake behavior.
- Conducted literature reviews and contributed to the interpretation, documentation, manuscript preparation, and presentation of project findings.

Graduate Coursework

- **Machine Learning and Data Science:** Machine Learning; Modern Statistical Methods for Big Data Analytics; Large-Scale Optimization; Genomic Data Science; Probability Theory for Data Science; Statistical Theory for Data Science
- **Numerical Analysis and Computational Modeling:** Scientific Computing in Applied Mathematics; Advanced Applied Numerical Methods; Numerical Solution of PDEs; Numerical and Computational Linear Algebra; Perturbation Methods
- **Mathematical Analysis:** Real Analysis; Complex Analysis; Measure Theory; Applied Functional Analysis; Fixed Point Theory; Engineering Analysis

Journal Publications

Metrics (October 2026): Citations: **29** | h-index: **3** | i10-index: **1** (Google Scholar)

1. **Yasin, N.**, Ahmad, S., Umair, M., Shah, Z., Vrinceanu, N., & Alhawael, G. (2025). Investigations of conjugate heat transfer and fluid flow in a partitioned porous cavity using the Darcy–Forchheimer model: Finite element-based computations. *Case Studies in Thermal Engineering*, *72*, 106252. DOI
2. Abid, M., **Yasin, N.**, Saqlain, M., Ul-Islam, S., & Ahmad, S. (2024). Numerical investigation and statistical analysis of the flow patterns behind square cylinders arranged in a staggered configuration utilizing the lattice Boltzmann method. *Journal of Applied Fluid Mechanics*, *17*(9), 1820–1843. DOI
3. Abid, M., Bibi, M., **Yasin, N.**, & Shahid, M. (2024). A novel computational analysis of boundary-driven two-dimensional heat flow with internal heat generation. *Computational Algorithms and Numerical Dimensions*, *3*(1), 1–16. DOI

Conference Presentations

1. **Yasin, N.** *Investigation of Flow Dynamics Around Bluff Bodies Using SRT-LBM*. Presented at MGO 2025, Syracuse University, Syracuse, NY, 2025.
2. **Yasin, N.** *Flow Structures Around Bluff Bodies at Low Reynolds Numbers*. Presented at the 15th Annual Math Awareness Conference, Old Dominion University, Norfolk, VA, 2025.
3. **Yasin, N.** *Flow Patterns Behind Stationary and Moving Bluff Bodies Utilizing the SRT-Lattice Boltzmann Method*. Presented at SEARCDE 2024, West Virginia University, Morgantown, WV, 2024.

Graduate Course Projects

- **Convergence and Performance Comparison of Numerical Solvers for Sparse Resistor-Network Linear Systems** 2023
  Used MATLAB to compare the convergence behavior and computational performance of numerical solvers for sparse linear systems arising from resistor-network models.
- **Image Processing and Compression Using Multiple Algorithms** Sep 2023 – Dec 2023
  Implemented image processing and compression techniques in MATLAB using SVD, FFT, DCT, and DWT, evaluating compression performance and image quality through compression ratio, MSE, and PSNR.

Academic Leadership and Service

- **President**, SIAM ODU Student Chapter, Old Dominion University — Sep 2024 – Present
- **Officer**, Mathematics and Statistics Club, Old Dominion University — Sep 2024 – Present
- **Organizer**, Math Awareness Conference, Old Dominion University — 2025 – 2026

Honors & Awards

- **KNU International Graduate Scholarship** 2021, 2022, 2023
  Kyungpook National University, South Korea
- **BK21 Plus Scholarship** 2022
  Kyungpook National University, South Korea
- **Prime Minister's National Laptop Scheme Award** 2018
  International Islamic University Islamabad, Pakistan
  Awarded based on academic merit.

Technical Skills

Programming and Parallel Computing: C++, CUDA, Fortran, Python, MATLAB
Numerical Methods: Lattice Boltzmann, Finite Difference, Finite Element, Finite Volume
Simulation and Visualization: COMSOL, ANSYS Fluent, ANSYS Workbench, Tecplot, Simulink
Development and Documentation: Visual Studio Code, Jupyter Notebook, Git, GitHub, LaTeX, Microsoft Office

Selected Training and Workshops

Deep Learning with MATLAB; Machine Learning with MATLAB; Python for Machine Learning; MATLAB for Math and Data Science

Languages

Pashto: Native **Urdu:** Advanced **English:** Advanced **Hindi:** Advanced **Punjabi:** Fluent